

The Hyperbolic Quartic Distribution

Probability Theory, Hyperbolic Laws of Large Numbers, and
Applications Across Economics, Finance, Strategy, and Computation

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Abstract

This paper introduces the Hyperbolic Quartic Distribution,

$$f(x) = \frac{3}{4} \operatorname{sech}^4(x),$$

a probability density generated from hyperbolic functions. We derive its normalization, cumulative distribution function, moment generating function, variance, and asymptotic properties. We establish ordinary and hyperbolic-coordinate Laws of Large Numbers, derive large-deviation principles, and discuss implications for economics, finance, political science, geopolitics, warfare economics, machine learning, artificial intelligence, data science, and strategic systems design. The distribution possesses exponential tails, bounded hyperbolic coordinates, and analytical tractability through Gamma-function representations.

The paper ends with “The End”

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1 Introduction

Probability distributions generated from trigonometric functions are common in statistics, while distributions generated from hyperbolic functions remain comparatively underexplored.

The hyperbolic secant distribution is known in probability theory. We generalize the idea by considering higher powers of the hyperbolic secant.

The central object of study is

$$f(x) = \frac{3}{4} \operatorname{sech}^4(x),$$

which we call the *Hyperbolic Quartic Distribution*.

The distribution combines:

- analytical tractability,
- exponential tail decay,
- bounded hyperbolic coordinates,
- finite moments of all orders,
- useful large-sample asymptotics.

2 Construction of the Distribution

Definition 1. *The Hyperbolic Quartic Distribution is defined by*

$$f(x) = \frac{3}{4} \operatorname{sech}^4(x), \quad x \in \mathbb{R}.$$

Since

$$\operatorname{sech}(x) = \frac{1}{\cosh(x)},$$

the density may also be written as

$$f(x) = \frac{3}{4 \cosh^4(x)}.$$

2.1 Normalization

Using

$$u = \tanh(x),$$

and

$$du = \operatorname{sech}^2(x) dx,$$

we obtain

$$\int_{-\infty}^{\infty} \operatorname{sech}^4(x) dx = \int_{-1}^1 (1 - u^2) du = \frac{4}{3}.$$

Hence

$$\int_{-\infty}^{\infty} f(x) dx = 1.$$

3 Distribution Function

The cumulative distribution function is

$$F(x) = \frac{3}{4} \int_{-\infty}^x \operatorname{sech}^4(t) dt.$$

Applying the substitution above gives

$$F(x) = \frac{1}{2} + \frac{3}{4} \tanh(x) - \frac{1}{4} \tanh^3(x).$$

Verification:

$$F(-\infty) = 0,$$

and

$$F(\infty) = 1.$$

4 Moments

Because the density is even,

$$E[X] = 0.$$

The variance is

$$\operatorname{Var}(X) = \frac{\pi^2 - 6}{9}.$$

Numerically,

$$\operatorname{Var}(X) \approx 0.429956.$$

4.1 Moment Generating Function

The moment generating function is

$$M(t) = \Gamma\left(2 + \frac{t}{2}\right) \Gamma\left(2 - \frac{t}{2}\right), \quad |t| < 4.$$

The cumulant generating function becomes

$$K(t) = \log M(t).$$

5 A General Hyperbolic Family

The quartic density belongs to a broader family.

Definition 2. For $n > 0$ define

$$f_n(x) = \frac{\Gamma(n + \frac{1}{2})}{\sqrt{\pi}\Gamma(n)} \operatorname{sech}^{2n}(x).$$

Parameter	Density
$n = \frac{1}{2}$	Hyperbolic secant distribution
$n = 1$	$\frac{1}{2} \operatorname{sech}^2(x)$
$n = 2$	$\frac{3}{4} \operatorname{sech}^4(x)$
$n = 3$	$\frac{15}{16} \operatorname{sech}^6(x)$

Table 1: Hyperbolic Density Family

As $n \rightarrow \infty$,

$$\operatorname{sech}^{2n}(x) \approx e^{-nx^2},$$

so suitably rescaled versions converge toward Gaussian distributions.

6 The Classical Law of Large Numbers

Let

$$X_1, X_2, \dots$$

be i.i.d. Hyperbolic Quartic random variables.

Define

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

Theorem 1 (Strong Law). For the Hyperbolic Quartic Distribution,

$$\bar{X}_n \xrightarrow{a.s.} 0.$$

Proof. The distribution possesses finite first moments.

The Kolmogorov Strong Law of Large Numbers therefore applies.

Since

$$E[X] = 0,$$

the result follows. □

7 Central Limit Theorem

Theorem 2. *The normalized sample mean satisfies*

$$\sqrt{n} \bar{X}_n \xrightarrow{d} N\left(0, \frac{\pi^2 - 6}{9}\right).$$

Consequently

$$\bar{X}_n = O_p(n^{-1/2}).$$

8 A Hyperbolic Law of Large Numbers

Define

$$U = \tanh(X).$$

Since

$$du = \operatorname{sech}^2(x) dx,$$

we obtain

$$g(u) = \frac{3}{4}(1 - u^2), \quad -1 < u < 1.$$

Theorem 3 (Hyperbolic LLN). *For*

$$U_i = \tanh(X_i),$$

we have

$$\frac{1}{n} \sum_{i=1}^n U_i \xrightarrow{a.s.} 0.$$

Proof. The transformed variable has bounded support.

Therefore all moments exist.

Applying the Strong Law yields

$$\frac{1}{n} \sum_{i=1}^n U_i \rightarrow E[U].$$

By symmetry,

$$E[U] = 0.$$

□

9 Large Deviations

The cumulant generating function

$$K(t) = \log \Gamma\left(2 + \frac{t}{2}\right) + \log \Gamma\left(2 - \frac{t}{2}\right)$$

produces the rate function

$$I(x) = \sup_t \{tx - K(t)\}.$$

Therefore

$$P\left(\frac{S_n}{n} \approx x\right) \asymp e^{-nI(x)}.$$

This provides exponential concentration analogous to classical Gaussian large-deviation theory.

10 Computational Algorithms

10.1 Sampling Algorithm

The inverse CDF does not possess a simple elementary form.

Rejection sampling is effective.

Pseudo-Code

Input: Desired sample size N

For $i = 1, \dots, N$:

 Repeat:

 Draw Y from logistic proposal

 Draw $U \sim \text{Uniform}(0,1)$

 Until

$U \leq f(Y)/(M g(Y))$

 Output Y

End

11 Machine Learning Applications

The bounded coordinate

$$u = \tanh(x)$$

creates natural robustness against outliers.

Potential applications include:

1. Robust neural network activations.
2. Hyperbolic embeddings.
3. Reinforcement learning reward shaping.
4. Bayesian priors with exponential tails.
5. Density estimation.
6. Generative modeling.

12 Data Science Implications

Compared to Gaussian assumptions:

Property	Gaussian	Hyperbolic Quartic
Support	Infinite	Infinite
Variance	Finite	Finite
Tail Type	Quadratic Exponent	Exponential
Bounded Hyperbolic Coordinate	No	Yes
Closed-form CDF	Partial	Yes

Table 2: Comparison with Gaussian Models

13 Economic Interpretation

Economic agents frequently exhibit bounded perception despite unbounded monetary quantities.

A transformation

$$u = \tanh(x)$$

captures diminishing sensitivity.

This resembles:

- prospect theory,
- bounded rationality,
- diminishing marginal utility,
- attention constraints.

The Hyperbolic LLN can therefore be interpreted as convergence of average perceived value.

14 Finance Applications

Potential uses include:

1. modeling bounded trader sentiment,
2. volatility forecasting,
3. risk aggregation,
4. stress testing,
5. utility-based portfolio optimization.

The exponential tails provide more realistic behavior than purely Gaussian assumptions while remaining mathematically tractable.

15 Political Science and International Relations

Political influence often saturates.

An increase from extreme power levels produces smaller marginal effects than an increase from moderate levels.

The mapping

$$u = \tanh(x)$$

naturally models such saturation.

Applications include:

- alliance dynamics,
- diplomatic influence,
- voting blocs,
- coalition formation.

16 Geopolitics and Warfare Economics

Military effectiveness often displays diminishing returns.

Examples include:

- force concentration,
- logistics capacity,
- information dominance,
- command-and-control complexity.

A hyperbolic coordinate representation models the reality that additional resources eventually produce bounded strategic effects.

The Hyperbolic LLN suggests convergence of aggregate strategic influence under repeated interactions.

17 Systems Theory Perspective

The transformation

$$u = \tanh(x)$$

compresses unbounded state spaces into bounded strategic coordinates.

This creates:

- numerical stability,
- bounded decision spaces,
- robust aggregation.

Such structures are valuable in cybernetics, control systems, and distributed computation.

18 Graph of the Density

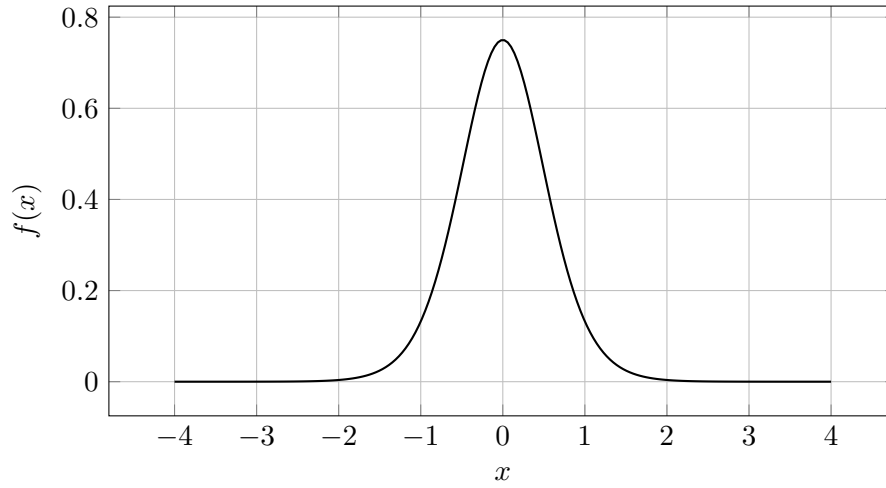


Figure 1: Hyperbolic Quartic Density

19 Conclusion

The Hyperbolic Quartic Distribution

$$f(x) = \frac{3}{4} \operatorname{sech}^4(x)$$

constitutes a tractable and analytically elegant probability law.

Its properties include:

- closed-form normalization,
- closed-form CDF,
- Gamma-function MGF,
- finite moments,
- classical LLN,
- hyperbolic-coordinate LLN,
- large-deviation principles,
- applications across economics, finance, political science, warfare economics, machine learning, and systems theory.

The broader family

$$f_n(x) = \frac{\Gamma(n + \frac{1}{2})}{\sqrt{\pi}\Gamma(n)} \operatorname{sech}^{2n}(x)$$

provides a fertile area for future research in probability, statistics, and complex adaptive systems.

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Glossary

Hyperbolic Quartic Distribution The density $\frac{3}{4} \operatorname{sech}^4(x)$.

Hyperbolic Coordinate The bounded variable $u = \tanh(x)$.

Strong Law Almost-sure convergence of sample means.

Large Deviations Exponential asymptotics of rare events.

Rate Function The Legendre transform of the cumulant generating function.

Bounded Rationality Decision-making under cognitive limitations.

Warfare Economics Economic analysis of military capability and strategic resources.

Hyperbolic LLN Law of Large Numbers applied in transformed hyperbolic coordinates.

The End